

HKUST MATH 2431 - problem set 8

2.

2. (*) Bertrand's paradox

[4 Points]

We consider an equilateral triangle inscribed in a circle of radius 1. Suppose a chord of the circle is chosen at random. What is the probability that the chord is longer than a side of the triangle?. For solving this try the following probability models:

- (a) The “random endpoints” method: Choose two uniform random points on the circumference of the circle and draw the chord joining them, i.e., let $U, V \sim \mathcal{U}((0, 1))$, define

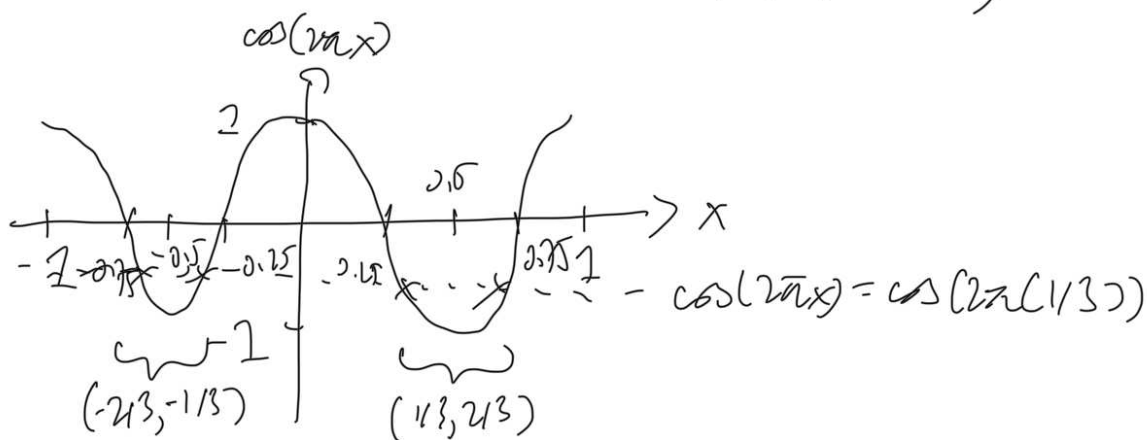
$$X = e^{U2\pi i} = (\cos(2\pi U), \sin(2\pi U)), \quad Y = e^{V2\pi i} = (\cos(2\pi V), \sin(2\pi V))$$

and take the chord connecting X and Y .

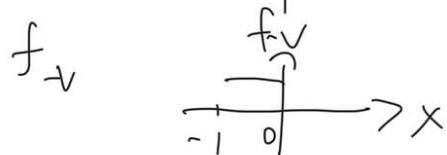
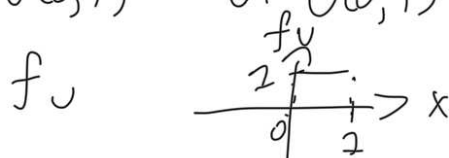
$$\begin{aligned}
 2.a. \quad L &= \|X - Y\| \\
 &= \|e^{U2\pi i} - e^{V2\pi i}\| \\
 &= \sqrt{(e^{U2\pi i} - e^{V2\pi i})(e^{-U2\pi i} - e^{-V2\pi i})} \\
 &= \sqrt{1 - e^{(U-V)2\pi i} - e^{-(U-V)2\pi i} + 1} \\
 &= \sqrt{2 - 2\cos(2\pi(U-V))}
 \end{aligned}$$

$$\begin{aligned}
 \text{side of equilateral } \triangle &= \sqrt{1^2 + 1^2 - 2\cos 120^\circ} \\
 &= \sqrt{2 - 2\cos(2\pi(1/3))}
 \end{aligned}$$

obviously to check that $\cos(2\pi(U-V)) < \cos(2\pi(1/3))$

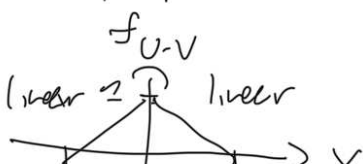


$$U, V \sim U(0, 1) \quad U \sim U(0, 1) \quad -V \sim U(-1, 0)$$



$$f_{U-V} = f_U * f_{-V}$$

(independent)

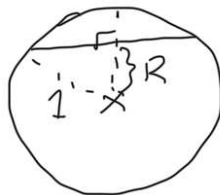


$$\text{req. prob.} = \frac{1}{2} \left(\left(\frac{1}{3} + \frac{2}{3} \right) \frac{1}{3} \right) \times 2 = \frac{1}{3}$$

This is also intuitively correct. Pick any point, then the other point must be $\leq 120^\circ$ away in both directions, so $\frac{1}{3}$.

- (b) The "random radius" method: Choose a radius of the circle, choose a uniform point on the radius and construct the chord through this point and perpendicular to the radius, i.e., choose a radius and choose $R \sim \mathcal{U}((0, 1))$, take the point on the radius that is at distance R from the center and the chord will be the only chord perpendicular to this point.

2.b. Take any radius. The scenario has radial symmetry, so this is okay. Consider the length



recall length of equi $\triangle$

$$= \sqrt{2 - 2\cos 120^\circ}$$

$$= \sqrt{2+1} = \sqrt{3}$$

$$L = 2\sqrt{1-R^2} = \sqrt{4-4R^2}$$

$$\sqrt{4-4R^2} \geq \sqrt{3} \Rightarrow R^2 \leq \frac{1}{4} \Rightarrow R \leq \frac{1}{2} \quad (R \geq 0)$$

so probability = $\frac{1}{2}$

- (c) The "random midpoint" method: Choose a point uniformly anywhere within the circle (see Problem 1.(a), but with A replaced by a disk) and construct a chord with the chosen point as its midpoint, i.e., and take the chord whose midpoint is (X, Y) .

2.c. define $R = \sqrt{x^2 + y^2}$

$$F_R(r) = \frac{\pi r^2}{\pi} = r^2 \quad (r \in [0, 1])$$

So we are reduced to (b) but with $R \sim F_R$.

eq. prob. = $F_R\left(\frac{1}{2}\right) - F_R(0) = \frac{1}{4}$

(d) Is this a contradiction?

2.d. No. These three methods parameterize the space of all possible chords differently, and then assigns uniform probability to the joint parameters. ^{almost everywhere}
These three different descriptions of the space of all possible chords while bijective, the transformations between them do not preserve area (Jacobian is not identity), so probability is not preserved. So the subspace for which the chord is longer than the equi Δ is assigned different probs by diff methods.
^{almost everywhere}

4. (*) **The student- t -distribution**

[4 Points]

Suppose that Z has the standard normal distribution, V has the χ^2 distribution with n degrees of freedom, and that Z and V are independent. The χ^2 -distribution with n degrees of freedom is defined as the distribution of $\sum_{i=1}^n X_i^2$ where $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Let

$$T = \frac{Z}{\sqrt{V/n}}.$$

You will show that T has p.d.f. given by

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2} \quad t \in \mathbb{R},$$

where we define $\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$, for $t > 0$.

- (a) Show that V has the probability density function $f_V(x) = c_n x^{n/2-1} e^{-x/2} \mathbb{1}_{\{x \geq 0\}}$, with an appropriate $c_n \in \mathbb{R}$.

Hint: Calculate c_n first, then use induction.

4.a. $V = \sum_{i=1}^n X_i^2$

$X_i: f_{X_i}(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ obv. $X_i^2 \geq 0$

$X_i^2: f_{X_i^2}(x) = \frac{1}{\sqrt{2\pi}} e^{-\sqrt{x}^2/2} (2)(\sqrt{x})' \mathbb{1}_{x \geq 0}(x)$
 pre-image of x^2
 $\{ \pm \sqrt{x}, \pm \sqrt{x} \}$
 $= \sqrt{\frac{2}{\pi}} e^{-x/2} \left(\frac{1}{2}\right) x^{-1/2} \mathbb{1}_{x \geq 0}(x)$
 $= \underbrace{\frac{1}{\sqrt{2\pi}}}_{C_1} \underbrace{x^{-1/2} e^{-x/2} \mathbb{1}_{x \geq 0}(x)}_{\text{matches } n=1}$

$C_1 = \frac{1}{\sqrt{2\pi}} \Rightarrow \frac{1}{C_1} = \sqrt{2\pi}$

$C_2 = \frac{1}{\int_0^\infty e^{-x/2} dx} = \frac{1}{[-2e^{-x/2}]_0^\infty} = \frac{1}{2} \Rightarrow \frac{1}{C_2} = 2$

$\frac{1}{C_n} = \int_0^\infty x^{n/2-1} e^{-x/2} dx = [-2x^{n/2-1} e^{-x/2}]_0^\infty + 2\left(\frac{n}{2}-1\right) \int_0^\infty x^{\frac{n}{2}-2} e^{-x/2} dx$
 $-2x^{n/2-1} e^{-x/2} \rightarrow 0$ as $x \rightarrow \infty$
 $= (0-0) + \frac{1}{C_{n-2}}$

$= \frac{n-2}{C_{n-2}}$

$\Rightarrow C_n = \frac{C_{n-2}}{n-2}$

$C_1 = \frac{1}{\sqrt{2\pi}}, C_2 = \frac{1}{2}, C_n = \frac{C_{n-2}}{n-2}$

Prove $f_{\sum_{i=1}^n X_i^2}(x) = C_n x^{\frac{n}{2}-1} e^{-x/2} \mathbb{1}_{x \geq 0}(x)$

for $n=1$, (proved above)

(assume $x \in [0, +\infty)$)

For $n=2$, $f_{V,2}(x) = (f_{V,1} * f_{V,1})(x)$ (independence)

C_n in terms of Γ $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$

$$c_1 = \frac{1}{\sqrt{\pi}}, c_2 = \frac{1}{2}, c_n = \frac{c_{n-2}}{n-2}$$

alternatively,
we

$$f_v(x) = c_n x^{\frac{n}{2}-1} e^{-\frac{x}{2}}$$

$$\frac{1}{c_n} = \int_0^\infty x^{\frac{n}{2}-1} e^{-\frac{x}{2}} dx$$

$$= 2 \int_0^\infty (2u)^{\frac{n}{2}-1} e^{-u} du$$

$$= 2^{\frac{n}{2}} \int_0^\infty u^{\frac{n}{2}-1} e^{-u} du$$

$$= 2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)$$

$$c_n = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)}$$

$$\frac{1}{2\pi} \int_0^x \tau^{\frac{n}{2}-1} e^{-\frac{\tau}{2}} (x-\tau)^{\frac{n}{2}-1} e^{-\frac{x-\tau}{2}} d\tau$$

$$= \frac{1}{2\pi} e^{-\frac{x}{2}} \int_0^x \tau^{\frac{n}{2}-1} (x-\tau)^{\frac{n}{2}-1} d\tau$$

$$= \frac{1}{\pi} e^{-\frac{x}{2}} \int_0^{\frac{x}{2}} \frac{1}{\sqrt{\left(\frac{x}{2}+\tau\right)\left(\frac{x}{2}-\tau\right)}} d\tau$$

$$= \frac{1}{\pi} e^{-\frac{x}{2}} \int_0^{\frac{x}{2}} \frac{1}{\sqrt{\frac{x^2}{4} - \tau^2}} d\tau$$

$$= \frac{1}{\pi} e^{-\frac{x}{2}} \frac{2}{x} \int_0^{\frac{x}{2}} \frac{1}{\sqrt{1 - \left(\frac{2}{x}\tau\right)^2}} d\tau$$

$$u = \frac{2}{x}\tau$$

$$= \frac{1}{\pi} e^{-\frac{x}{2}} \int_0^1 \frac{1}{\sqrt{1-u^2}} du$$

$$= \frac{1}{\pi} e^{-\frac{x}{2}} [\arcsin(u)]_0^1$$

$$= \frac{1}{2} e^{-\frac{x}{2}} \quad (x \in [0, +\infty))$$

So proved for $n=2$ matches c_n when $n=2$

Now assume $f_{v,n}(x) = c_n x^{\frac{n}{2}-1} e^{-\frac{x}{2}} \mathbb{1}_{x \geq 0}(x)$

$$f_{v,n+2}(x) = (f_{v,n} * f_{v,2})(x) \quad (\text{assume } x \in [0, +\infty))$$

$$(\text{independence}) = \frac{c_n}{2} \int_0^x \tau^{\frac{n}{2}-1} e^{-\frac{\tau}{2}} e^{-\frac{x-\tau}{2}} d\tau$$

$$= \frac{c_n}{2} e^{-\frac{x}{2}} \int_0^x \tau^{\frac{n}{2}-1} d\tau$$

$$= \frac{c_n}{2\left(\frac{n+2}{2}-1\right)} e^{-\frac{x}{2}} \left[\tau^{\frac{n+2}{2}-1}\right]_0^x$$

$$= \frac{c_n}{n} e^{-\frac{x}{2}} x^{\frac{n+2}{2}-1} \quad (x \in [0, +\infty))$$

$$C_{n+2} = \frac{C_n}{n} \quad \text{matches the case for } n+2$$

So proved by induction for all $n \in \mathbb{N}_{\neq 0}$

(b) Find the joint probability density function of (T, V) .

Hint: As always, look at the (joint) cumulative distribution function $F_{T,V}(t, v)$.

$$4.3. \quad T = \frac{Z}{\sqrt{V/n}} \quad F_{T,V}(t, v) = P(T \leq t, V \leq v)$$

$$\text{Consider } T' = \frac{1}{\sqrt{V/n}} = \sqrt{\frac{n}{V}}$$

$$\text{Then } f_{T',V}(t, v) = C_n v^{\frac{n}{2}-1} e^{-\frac{v}{2}} \delta(t - \sqrt{\frac{n}{v}})$$

↳ Dirac delta

$$Z \sim N(0, 1)$$

$$\text{a } Z \sim N(0, a^2) \quad f_{aZ}(x) = \frac{1}{\sqrt{2\pi}a} \exp\left(-\frac{x^2}{2a^2}\right)$$

$$\because T = ZT', \text{ and } Z \perp T' \Leftrightarrow Z \perp V$$

$$\text{So } f_{T,V}(t, v) = C_n v^{\frac{n}{2}-1} e^{-\frac{v}{2}} \frac{1}{\sqrt{2\pi n/v}} \exp\left(-\frac{t^2}{2n}\right)$$

$$\left(C_n \text{ is intentionally kept here, see above for } C_n \text{ in terms of } \Gamma \right) = \frac{C_n}{\sqrt{2\pi n}} v^{\frac{n}{2}} \exp\left(-\frac{v}{2} \frac{n+t^2}{n}\right)$$

(c) Find the probability density function of T .

$$f_T(t) = \int_0^{+\infty} f_{T,v}(t,v) dv \quad v \geq 0$$

$$= \frac{C_n}{\sqrt{2\pi n}} \int_0^{+\infty} v^{\frac{n-1}{2}} \exp\left(-\frac{v}{2} \left(1 + \frac{t^2}{n}\right)\right) dv$$

$$u := \frac{v}{2} \left(1 + \frac{t^2}{n}\right)$$

$$du = \frac{1}{2} \left(1 + \frac{t^2}{n}\right) dv$$

$$v = \frac{2u}{1 + \frac{t^2}{n}}$$

$$= \frac{C_n}{\sqrt{2\pi n}} \frac{2}{1 + \frac{t^2}{n}} \int_0^{+\infty} \left(\frac{2u}{1 + \frac{t^2}{n}}\right)^{\frac{n-1}{2}} \exp(-u) du$$

$$= \frac{C_n}{\sqrt{2\pi n}} \left(\frac{2}{1 + \frac{t^2}{n}}\right)^{\frac{n-1}{2}} \int_0^{+\infty} u^{\frac{n-1}{2}} \exp(-u) du$$

$$= \frac{\Gamma((n+1)/2) 2^{n/2} C_n}{\sqrt{2\pi n}} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}$$

$$= \frac{\Gamma((n+1)/2)}{\sqrt{\pi n} \Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}$$

from above

$$C_n = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})}$$